

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination – 2023

Course: B. Tech. Branch : Information Technology Semester :VII

Subject Code & Name: Machine Learning (BTITOE704B)

Max Marks: 60

Date: 09/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		
A) Explain the well-posed learning problems for designing a machine learning system in detail with some examples.	CO1	6
B) Explain the concept of cross validation in detail.	CO2	6
C) What is hypothesis testing? Explain the hypothesis testing with respect to a machine learning system with example.	CO2	6
Q.2 Solve Any Two of the following.		
A) Explain linear regression & logistic regression algorithms with at least 2 problems for each algorithm where you will use these algorithms.	CO4	6
B) Explain the following terms. Bias, Variance, Training & Testing, Overfitting, Occam's razor, Pruning	CO1	6
C) Explain the Bayes classifier with example.	CO4	6
Q.3 Solve Any Two of the following.		
A) Explain the architecture and working of Multilayer Perceptrons (MLPs) in detail.	CO5	6
B) Explain the backpropagation algorithm of neural network in detail.	CO5	6
C) Explain various activation functions in detail.	CO7	6
Q.4 Solve Any Two of the following.		
A) Explain the working and use of principal component analysis.	CO7	6
B) Write a short note on autoencoders.	CO4	6
C) Explain the concept of regularization in detail.	CO5	6
Q.5 Solve Any Two of the following.		
A) Explain vanishing and exploding gradients problems in detail.	CO5	6
B) Explain various functional layers in CNN in detail.	CO4	6
C) Explain working and application of recurrent neural networks.	CO7	6